

Triangle groups and tilings

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Drawn with the **luahyperbolic** package. (<https://github.com/dmegy/luahyperbolic>)

The picture is a reproduction of fig.1 p. 442 in Parker, Wang, Xie, *Complex Hyperbolic $(3, 3, n)$ Triangle Groups*, Pacific Journal of Math., vol 289 no. 2 (2016), available at <https://msp.org/pjm/2016/280-2/pjm-v280-n2-p08-p.pdf>.

```
\begin{luacode*}
local PI = math.pi
local A, B, C = hyper.triangleWithAngles(PI/3, PI/3, PI/5)
local polygon = {}
for k=1,10 do
    table.insert(polygon, hyper.rotation(A,k*PI/5)(C))
end
hyper.tikzBegin("scale=4,rotate=190")
for i=1,10 do
    hyper.drawSegment(A, polygon[i])
    hyper.drawSegment(polygon[i], polygon[i%10+1], "very thick")
end
for i=1,4 do
    local P,Q = polygon[i],polygon[i+1]
    hyper.drawLine(P,Q)
    hyper.labelPoint((P+Q)/1.8, "$\\mathcal{B}_{(i-2)..(i-1)}$", "")
    hyper.labelPoint((P+Q)/6.5, "$n$", "")
    hyper.labelPoint((5*P+Q)/6.8, "$3$", "")
    hyper.labelPoint((P+5*Q)/6.8, "$3$", "")
end
hyper.tikzEnd()
\end{luacode*}
```

